CS 171 / EE 142: Intro to ML and DM

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UC Riverside

Slide Set 5: Perceptron

Slides from CS 171 / EE 142

- From UC Riverside
  - CS 171 / EE 142: Introduction to Machine Learning and Data Mining
  - Professor Christian Shelton
- DO NOT REDISTRIBUTE
  - For use only by enrolled students in the course

Binary Classification

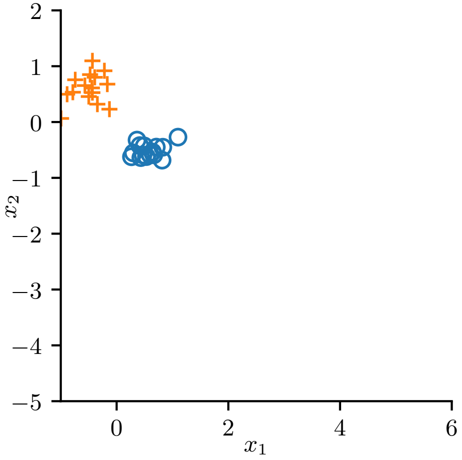
Let $y_i \in \{-1, +1\}$

Binary Classification

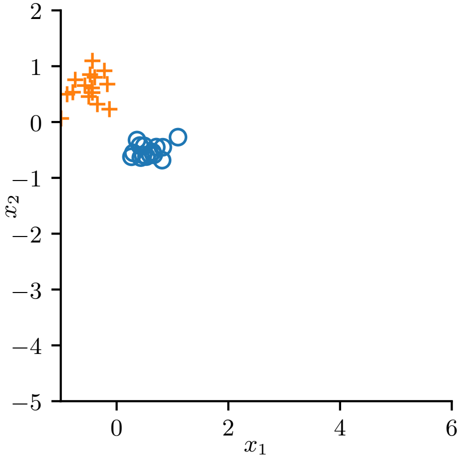
Let $y_i \in \{-1, +1\}$

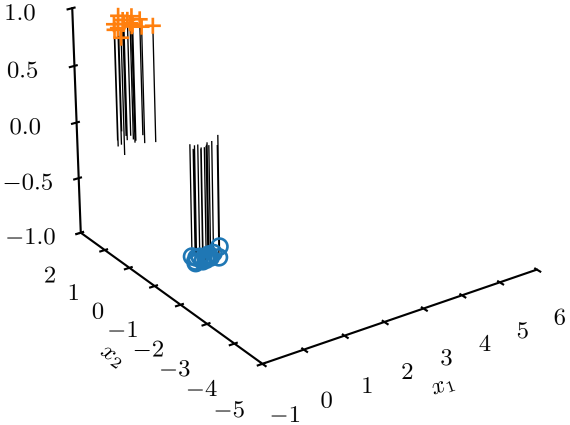
Why not just use LLS?
If $f(\vec{x}) > 0$, report class "+1" Otherwise, report class "-1"

LLS for Classification



LLS for Classification





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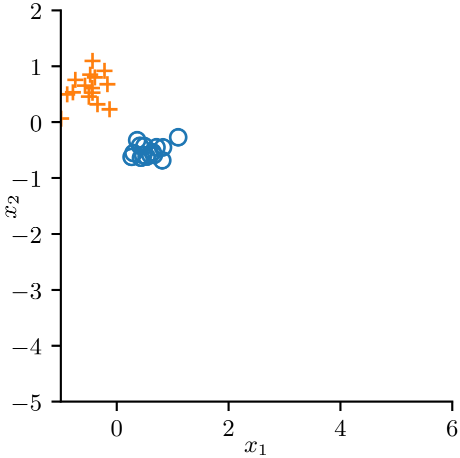

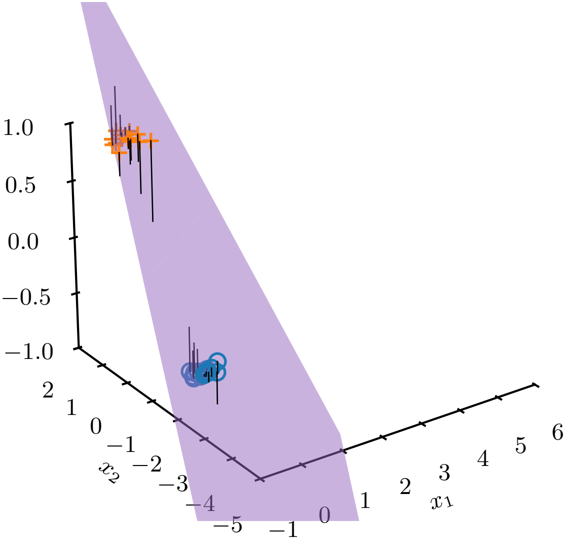

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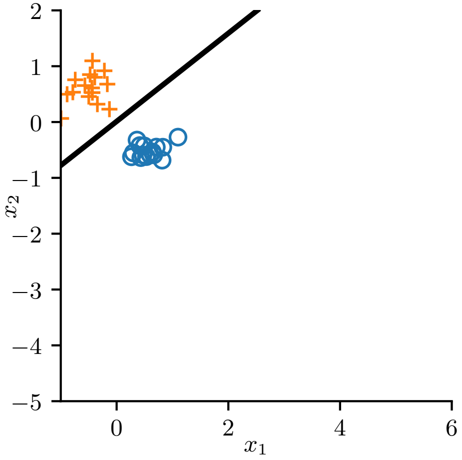

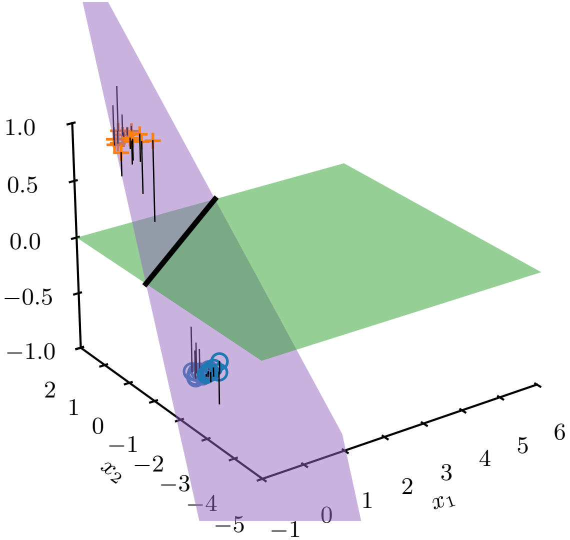

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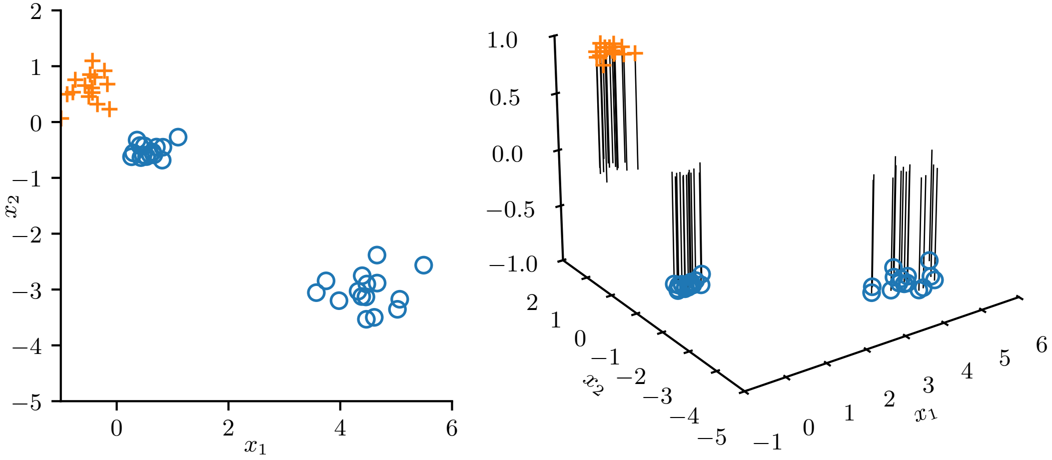
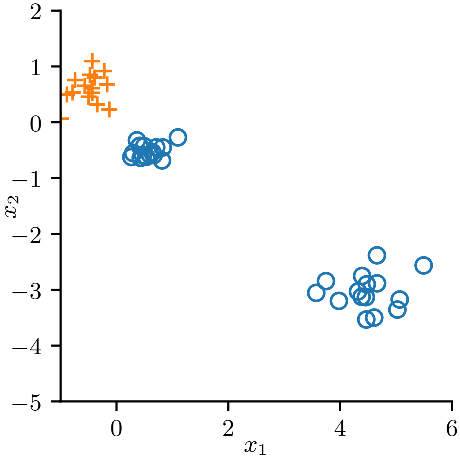
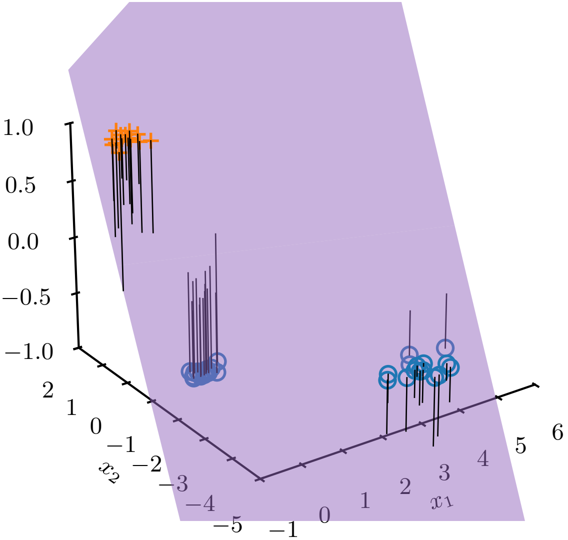


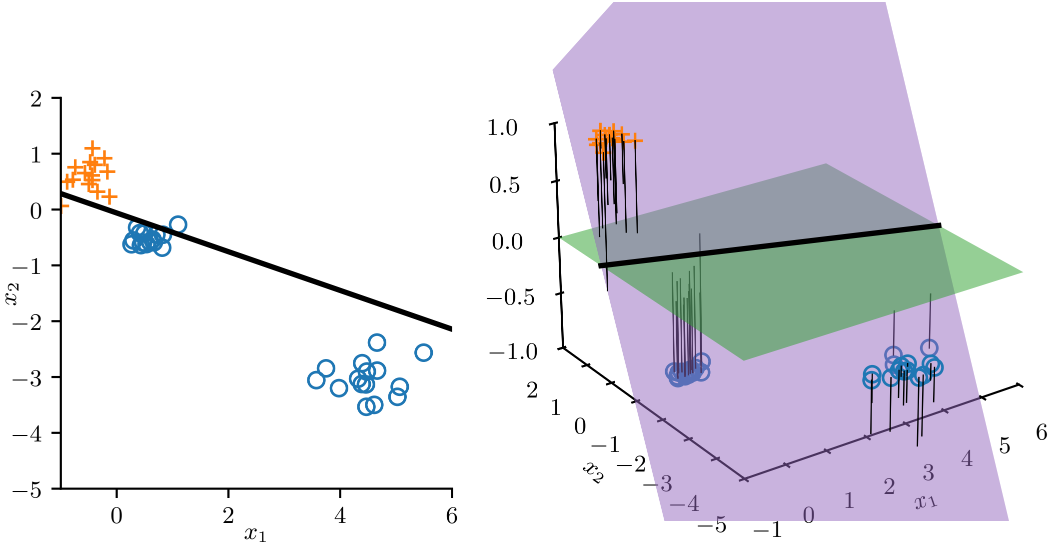


2
1
0
−1
−2
−3
−4
−5
0
2
4
6
x_1
x_2

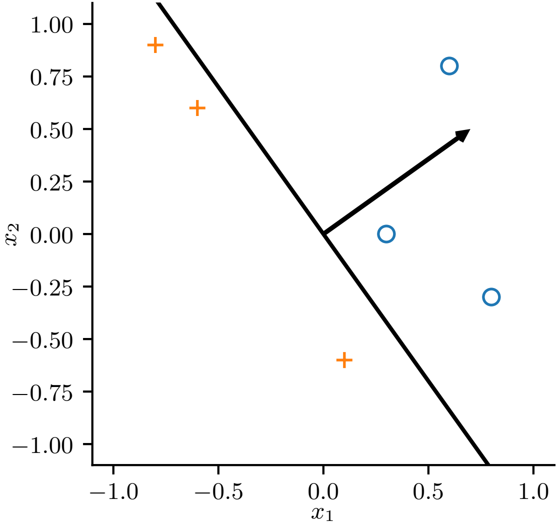


1.0
0.5
0.0
−0.5
−1.0
2
1
0
−1
−2
−3
−4
−5
x_2
−1
0
1
2
3
4
5
6
x_1

LLS for Classification

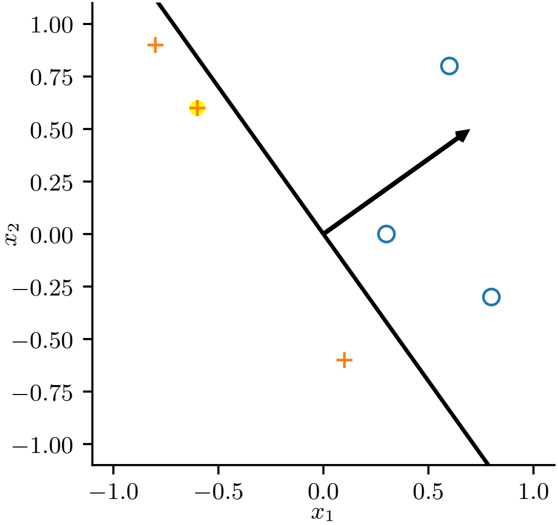


Perceptron Learning Algorithm, Example



$$\vec{w} = \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix}$$

Perceptron Learning Algorithm, Example

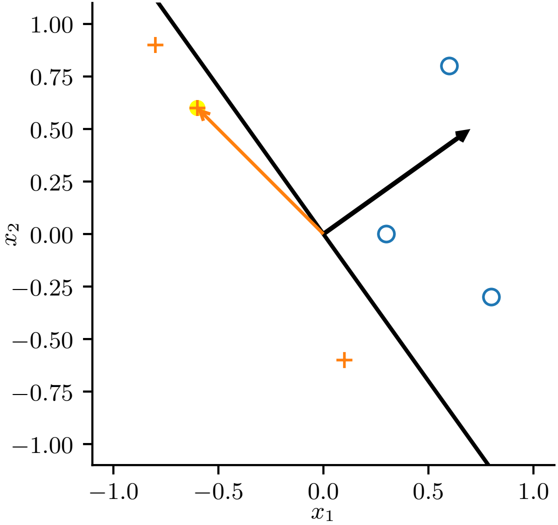


$$\vec{w} = \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} -0.6 \\ 0.6 \end{bmatrix}$$

$$y_i = +1$$

Perceptron Learning Algorithm, Example



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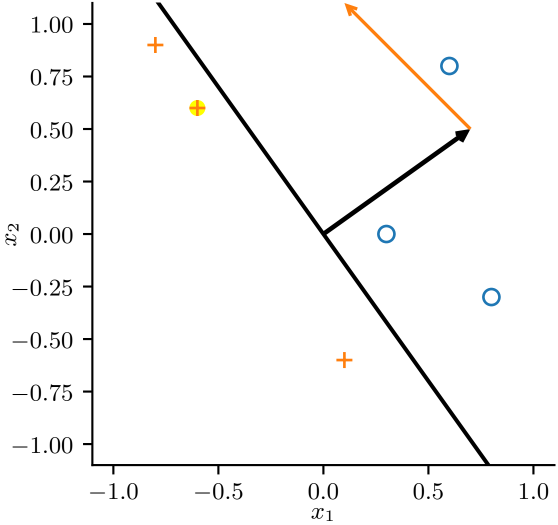
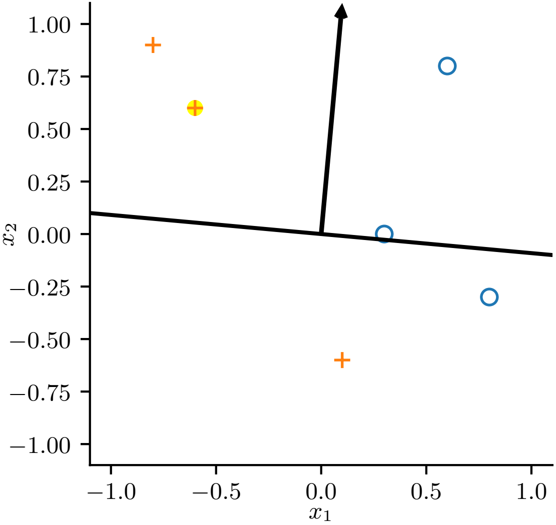


$$\vec{w} = \begin{bmatrix} 0.7 - 0.6 \\ 0.5 + 0.6 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} -0.6 \\ 0.6 \end{bmatrix}$$

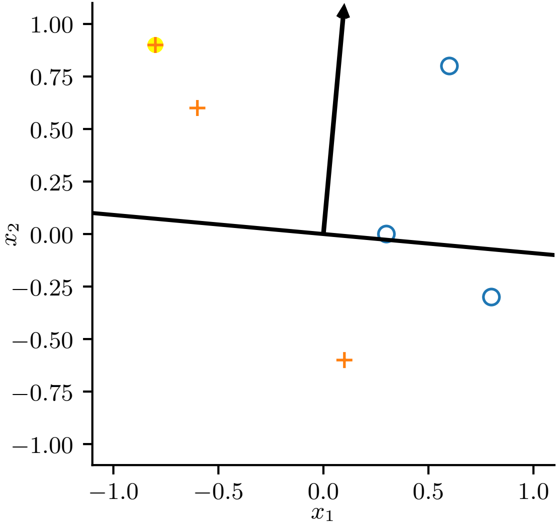
$$y_i = +1$$

Perceptron Learning Algorithm, Example



$$\vec{w} = \begin{bmatrix} 0.1 \\ 1.1 \end{bmatrix}$$

Perceptron Learning Algorithm, Example

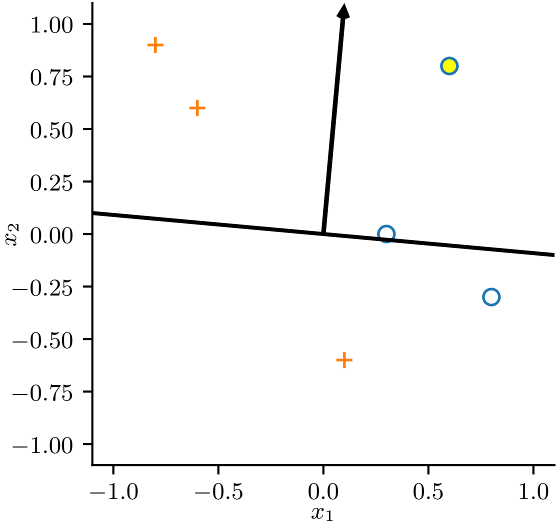


$$\vec{w} = \begin{bmatrix} 0.1 \\ 1.1 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} -0.8 \\ 0.9 \end{bmatrix}$$

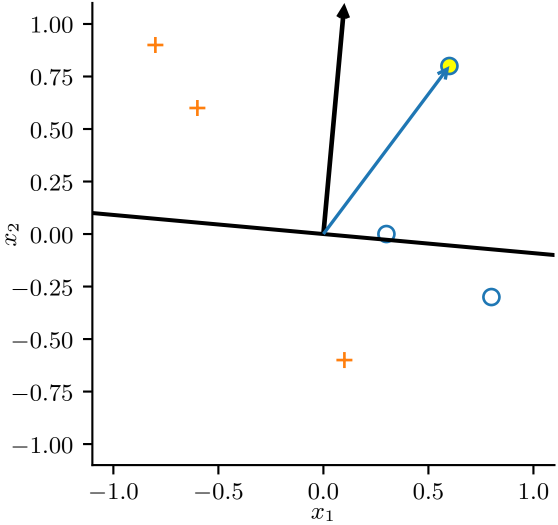
$$y_i = +1$$

Perceptron Learning Algorithm, Example



$$\vec{w} = \begin{bmatrix} 0.1 \\ 1.1 \end{bmatrix}$$

Perceptron Learning Algorithm, Example

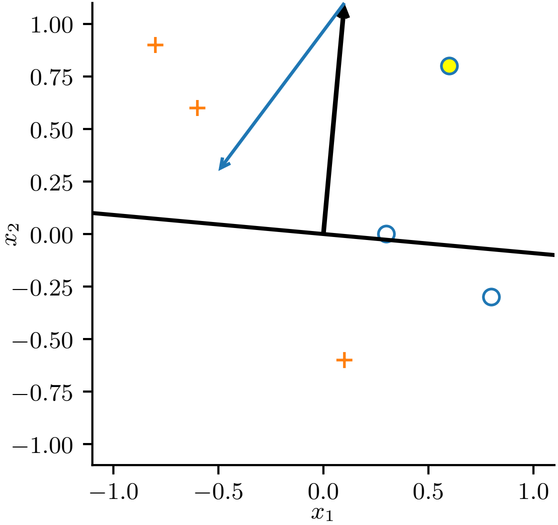


$$\vec{w} = \begin{bmatrix} 0.1 \\ 1.1 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

$$y_i = -1$$

Perceptron Learning Algorithm, Example

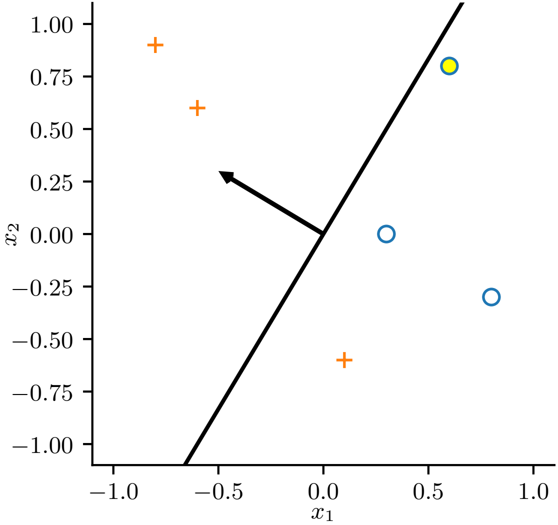


$$\vec{w} = \begin{bmatrix} 0.1 - 0.6 \\ 1.1 - 0.8 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

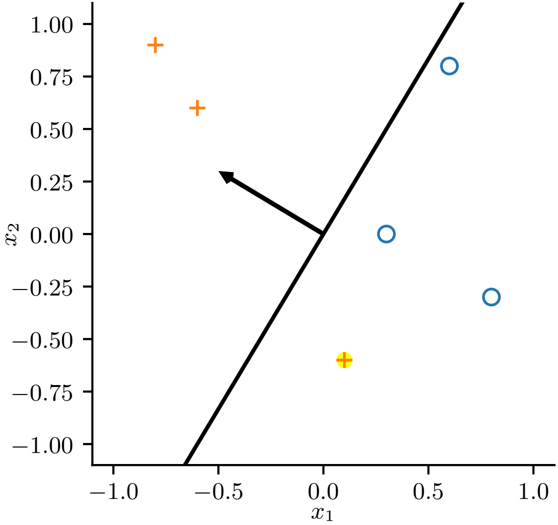
$$y_i = -1$$

Perceptron Learning Algorithm, Example



$$\vec{w} = \begin{bmatrix} -0.5 \\ 0.3 \end{bmatrix}$$

Perceptron Learning Algorithm, Example

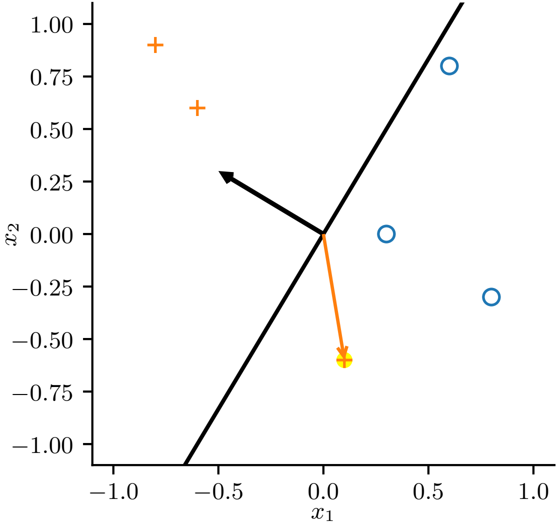


$$\vec{w} = \begin{bmatrix} -0.5 \\ 0.3 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} 0.1 \\ -0.6 \end{bmatrix}$$

$$y_i = +1$$

Perceptron Learning Algorithm, Example



$$\vec{w} = \begin{bmatrix} -0.5 \\ 0.3 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} 0.1 \\ -0.6 \end{bmatrix}$$

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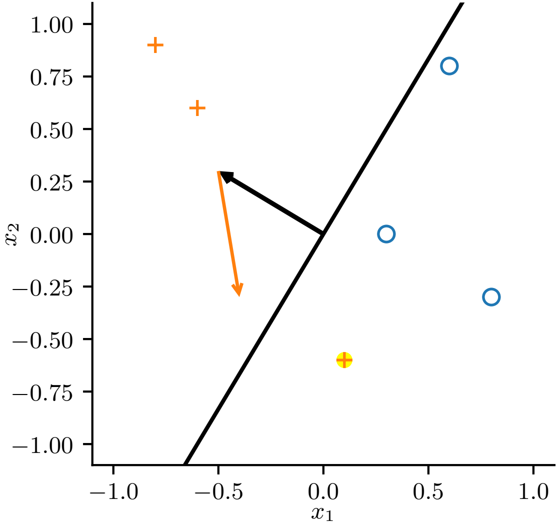
Perceptron Learning Algorithm, Example

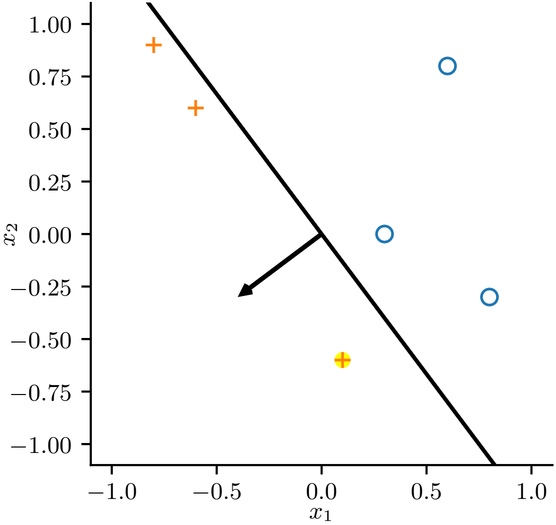


$$\vec{w} = \begin{bmatrix} -0.5 + 0.1 \\ 0.3 - 0.6 \end{bmatrix}$$

$$\vec{x}_i = \begin{bmatrix} 0.1 \\ -0.6 \end{bmatrix}$$

$$y_i = +1$$

Perceptron Learning Algorithm, Example



$$\vec{w} = \begin{bmatrix} -0.4 \\ -0.3 \end{bmatrix}$$

Perceptron Learning Algorithm

1. Let w be an arbitrary starting vector
2. Let η be a positive scalar (usually $\eta = 1$)
3. While not done
    1. For $i = 1, \ldots, m$
        1. $h_i \leftarrow \vec{w}^\top \vec{x}_i$
        2. If $h_i y_i < 0$
           Then $\vec{w} \leftarrow \vec{w} + \eta y_i \vec{x}_i$

Perceptron Convergence

Perceptron Convergence Theorem:
If the data are linearly separable,
the perceptron learning algorithm will find a separating hyperplane in a finite number of steps.
If the data are not linearly separable,
it will run forever.

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If the data are linearly separable,
the perceptron learning algorithm will find a separating hyperplane in a finite number of steps.
If the data are not linearly separable,
it will run forever.

No (easy) way to tell the difference unless it stops.

(Linear) Discriminants

The learned function is

$$f(\vec{x}) = \operatorname{sgn}(\vec{x}^{\mathsf{T}}\vec{w}) = \begin{cases} +1, & \text{if } \vec{x}^{\mathsf{T}}\vec{w} > 0 \\ -1, & \text{otherwise} \end{cases}$$

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$\vec{x}^{\mathsf{T}}\vec{w}$ is known as the **discriminant**

We will denote the discriminant function as $\widetilde{f}$:

$$f(\vec{x}) = \mathrm{sgn}(\widetilde{f}(\vec{x}))$$
$$\widetilde{f}(\vec{x}) = \vec{x}^{\mathsf{T}}\vec{w}$$

(Linear) Discriminants

The learned function is

$$f(\vec{x}) = \operatorname{sgn}(\vec{x}^\intercal \vec{w}) = \begin{cases} +1, & \text{if } \vec{x}^\intercal \vec{w} > 0 \\ -1, & \text{otherwise} \end{cases}$$

$\vec{x}^\intercal \vec{w}$ is known as the **discriminant**

We will denote the discriminant function as $\widetilde{f}$:

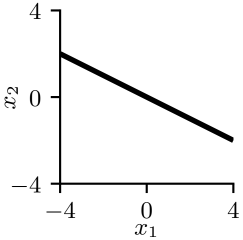
$$f(\vec{x}) = \operatorname{sgn}(\widetilde{f}(\vec{x}))$$
$$\widetilde{f}(\vec{x}) = \vec{x}^\intercal \vec{w}$$

Because

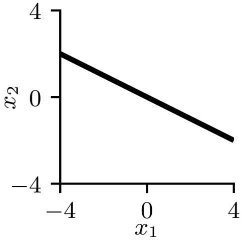
- $y = \widetilde{f}(\vec{x})$ is linear,
- $y = 0$ is linear, and
- the intersection of two linear surfaces is linear

the **descision boundary** is linear.

Linear Decision Boundaries

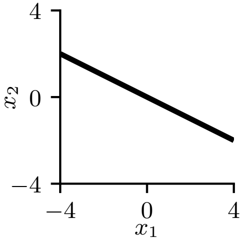


Linear Decision Boundaries



$$x_2 = -0.5x_1$$

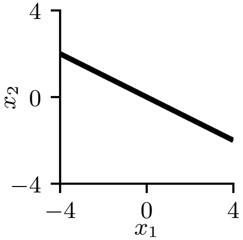
Linear Decision Boundaries



$$x_2 = -0.5x_1$$

$$x_1 = -2x_2$$

Linear Decision Boundaries



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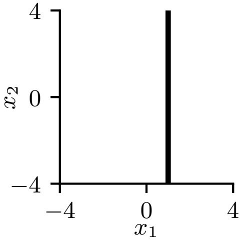

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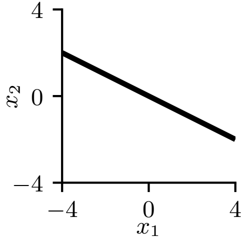

$$x_2 = -0.5x_1$$
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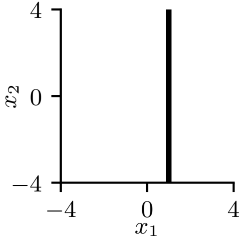
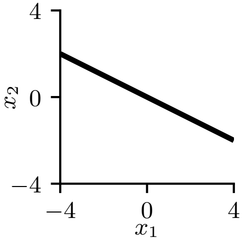


$$x_1 = 1$$

Linear Decision Boundaries



$$x_2 = -0.5x_1$$

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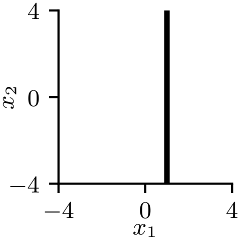
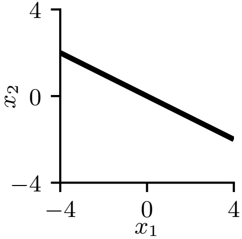


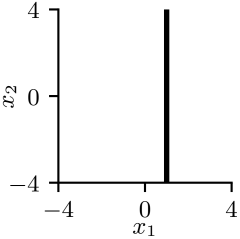
$x_2 =$??

$$x_1 = 1$$

Linear Decision Boundaries

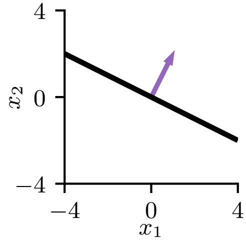


$$x_2 = -0.5x_1$$
$$x_1 = -2x_2$$
$$0 = x_1 + 2x_2$$

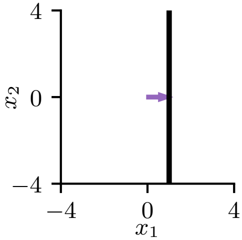


$$x_2 = ??$$
$$x_1 = 1$$
$$0 = -1 + x_1$$

Linear Decision Boundaries



$$x_2 = -0.5x_1$$
$$x_1 = -2x_2$$
$$0 = x_1 + 2x_2$$
$$0 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}^\mathsf{T} \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$



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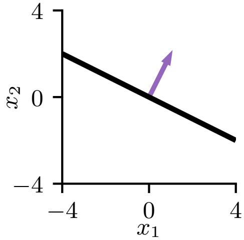
Linear Decision Boundaries



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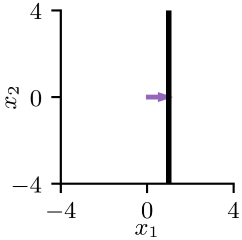

$$x_2 = ??$$
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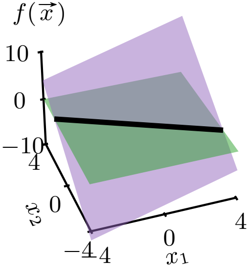
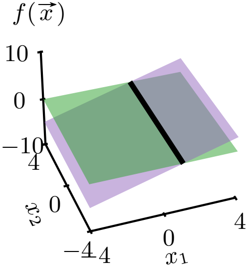
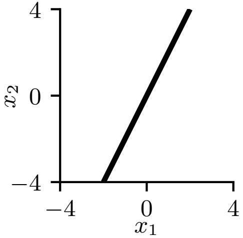


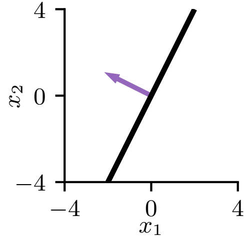


Linear Decision Boundaries

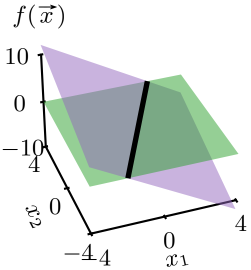


$\vec{w} =$

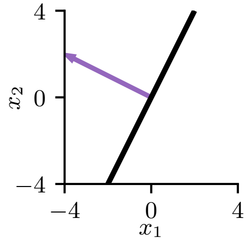
Linear Decision Boundaries



$$\vec{w} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

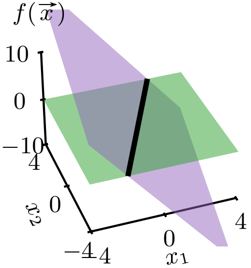


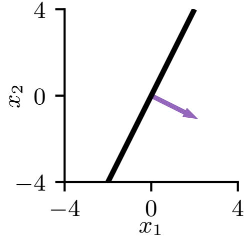
Linear Decision Boundaries



$$\vec{w} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ -4 \\ 2 \end{bmatrix}$$



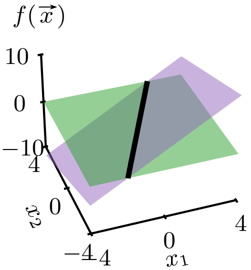


$$\vec{w} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

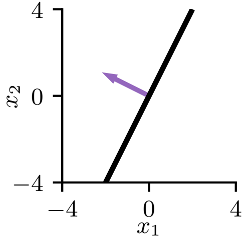
$$\vec{w} = \begin{bmatrix} 0 \\ -4 \\ 2 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$



Linear Decision Boundaries



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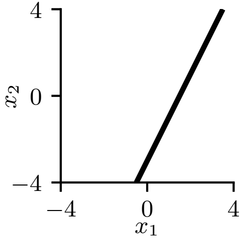
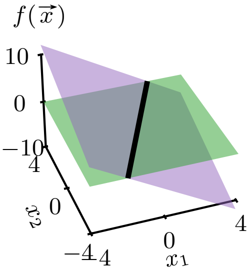
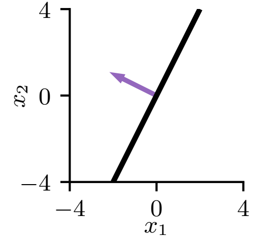
$$\vec{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

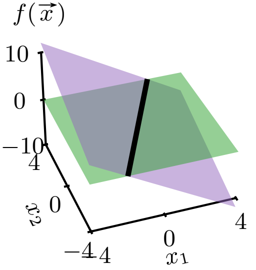
$$\vec{w} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$





Linear Decision Boundaries



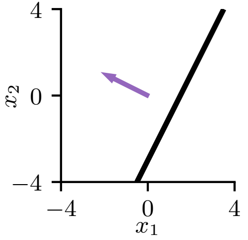


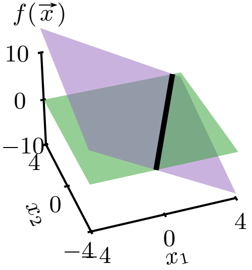
$$\vec{w} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ -4 \\ 2 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

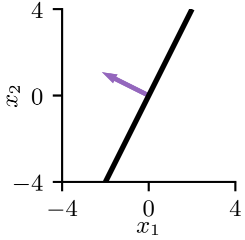
$$\vec{w} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$





$$\vec{w} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

Linear Decision Boundaries

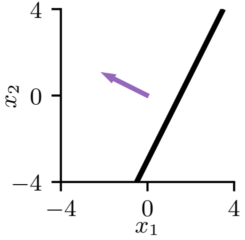


$$\vec{w} = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ -4 \\ 2 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

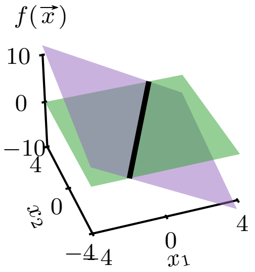


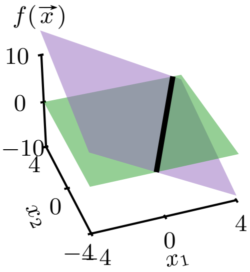
$$\vec{w} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 6 \\ -4 \\ 2 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} -6 \\ 4 \\ -2 \end{bmatrix}$$





Problems with Perceptrons

- Does not distinguish between different separating lines/planes/hyper-planes
- Does not work when data are not separable
- $\vec{w}^\mathsf{T}\vec{x}$ is treated the same as $(2\vec{w})^\mathsf{T}\vec{x}$
  - Difficult if used for multi-class
  - Why the extra free parameter?

Solution?

Problems with Perceptrons

- Does not distinguish between different separating lines/planes/hyper-planes
- Does not work when data are not separable
- $\vec{w}^\top \vec{x}$ is treated the same as $(2\vec{w})^\top \vec{x}$
  - Difficult if used for multi-class
  - Why the extra free parameter?

Solution?
Make the discriminant's magnitude (not just sign) correspond to something.